OS- ORANGE- U4

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SRN: PES2UG24CS082
GitHub Link: https://github.com/Gowda-Arjun/PES2UG24CS082-pes-vcs

Screenshots:

1A

```
✗ Fri 17 Apr - 22:01 ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 4●
@cowssaymoo ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B

```
Fri 17 Apr - 22:01 ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 4●
@cowssaymoo find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2A

```
Fri 17 Apr - 22:08  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑3 1* 4●
@cowssaymoo  make test_tree
make: 'test_tree' is up to date.

Fri 17 Apr - 22:10  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 3●
@cowssaymoo  ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B

```
Fri 17 Apr - 22:16  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑2 1* 3●
@cowssaymoo  find .pes/objects -type f

Fri 17 Apr - 22:17  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑2 1* 3●
@cowssaymoo  xxd .pes/objects/XX/YYY... | head -20
xxd: .pes/objects/XX/YYY...: No such file or directory
```

3A

```
Fri 17 Apr - 22:37  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 2●
@cowssaymoo  ./pes init
Initialized empty PES repository in .pes/

Fri 17 Apr - 22:37  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 2●
@cowssaymoo  echo "hello" > file1.txt

Fri 17 Apr - 22:37  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 1* 2●
@cowssaymoo  echo "helloooo" > file2.txt

Fri 17 Apr - 22:37  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 2* 2●
@cowssaymoo  ./pes add file1.txt file2.txt

Fri 17 Apr - 22:38  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 2* 2●
@cowssaymoo  ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_sequence.sh
  untracked:  README.md
  untracked:  .gitignore
  untracked:  test_objects
  untracked:  tree.c
  untracked:  test_objects.c
  untracked:  commit.c
  untracked:  commit.h
  untracked:  test_tree
  untracked:  pes.c
  untracked:  pes.h
  untracked:  index.c
  untracked:  Makefile
  untracked:  test_tree.c
  untracked:  object.c
  untracked:  index.h
```

3B

```
Fri 17 Apr - 22:38  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 2* 2●
@cowssaymoo  cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776445663 6 file1.txt
100644 de268ff198bb3a61e935aa8bff7c6beb17935742cf5479305bb65cf4312aa6fc 1776445678 9 file2.txt
```

4A

```
@cowssaymoo  ./pes add file1.txt

Fri 17 Apr - 22:49  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2* 1●
@cowssaymoo  ./pes log
No commits yet.

Fri 17 Apr - 22:49  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2* 1●
@cowssaymoo  ./pes commit -m "first commit"
Committed: 5dd00a7861c4... first commit

Fri 17 Apr - 22:50  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2* 1●
@cowssaymoo  ./pes log
commit 5dd00a7861c44153f0f11be9f98952bb5b08f328b2a2c7744be71958703dac35
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776446426

    first commit


Fri 17 Apr - 22:50  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2* 1●
@cowssaymoo  ./pes add file2.txt

Fri 17 Apr - 22:50  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2* 1●
@cowssaymoo  ./pes commit -m "second commit"
Committed: ed7ad8d1b5d2... second commit

Fri 17 Apr - 22:50  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2* 1●
@cowssaymoo  ./pes log
commit ed7ad8d1b5d2284651f72866e7f88e1bf93d1b4385dd71114f9426f4fbba5802
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776446455

    second commit

commit 5dd00a7861c44153f0f11be9f98952bb5b08f328b2a2c7744be71958703dac35
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776446426

    first commitirst commiA
```

4B

```
Fri 17 Apr - 22:55  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 3* 1●
@cowssaymoo  find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/5d/d00a7861c44153f0f11be9f98952bb5b08f328b2a2c7744be71958703dac35
.pes/objects/7f/b55cd7288f7d0f495a2038305076cecd966e1cccefb71e17100ddd17fc1762
.pes/objects/a0/db10380726812acf14c429d2b52ff3a7e44f81e704b527bc5a610feac03807
.pes/objects/de/268ff198bb3a61e935aa8bff7c6beb17935742cf5479305bb65cf4312aa6fc
.pes/objects/ed/7ad8d1b5d2284651f72866e7f88e1bf93d1b4385dd71114f9426f4fbba5802
.pes/refs/heads/main
```

4C

```
Fri 17 Apr - 22:55  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑1 3* 1●
@cowssaymoo  cat .pes/refs/heads/main
a0db10380726812acf14c429d2b52ff3a7e44f81e704b527bc5a610feac03807

Fri 17 Apr - 23:00  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main ↑3 2* 1●
@cowssaymoo  cat .pes/HEAD
ref: refs/heads/main
```

FINAL:

```
Fri 17 Apr - 23:21  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2*
@cowssaymoo  make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes

Fri 17 Apr - 23:21  ~/gitstuff/PES2UG24CS082-pes-vcs  origin main 2*
@cowssaymoo  make test-integration
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: f502277f4bcf... Initial commit

Log after first commit:
commit f502277f4bcffffbcf642b8de9b12cb7363c14fa95cc6192efe2e29eedb0495a
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776448285

    Initial commit


--- Second Commit ---
Committed: afab4e23ea70... Update file.txt

--- Third Commit ---
Committed: 0b1c9a3f9498... Add farewell
```

```
--- Second Commit ---
Committed: afab4e23ea70... Update file.txt

--- Third Commit ---
Committed: 0b1c9a3f9498... Add farewell

--- Full History ---
commit 0b1c9a3f9498afa13fbb49c506cfe865853f5e93df2516c4464803a38136f7b9
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776448285

    Add farewell

commit afab4e23ea705fb393dc3f951609f7c422bd1c5a05504ae2717ee27f2351c9dd
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776448285

    Update file.txtte file.txt

commit f502277f4bcffffbcf642b8de9b12cb7363c14fa95cc6192efe2e29eedb0495a
Author: Arjun Mohan Gowda <PES2UG24CS082>
Date:   1776448285

    Initial committial comA


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
0b1c9a3f9498afa13fbb49c506cfe865853f5e93df2516c4464803a38136f7b9

--- Object Store ---
Objects created:
10
.pes/objects/0b/1c9a3f9498afa13fbb49c506cfe865853f5e93df2516c4464803a38136f7b9
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/af/ab4e23ea705fb393dc3f951609f7c422bd1c5a05504ae2717ee27f2351c9dd
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/f5/02277f4bcffffbcf642b8de9b12cb7363c14fa95cc6192efe2e29eedb0495a

=== All integration tests completed ===
```

PHASE 5 AND 6:

Phase 5 & 6: Analysis Questions

Q5.1: Checkout implementation and why it is complex
Checkout must update three states in a safe order: repository metadata, index, and working directory.
Metadata flow:
1. Resolve target branch name to its commit hash using branch refs.
2. Read current commit from HEAD (directly or via current branch).
3. Keep HEAD as a branch reference (if checking out a branch), but move it to the target branch.
4. Update the index to match the target commit's tree after files are successfully updated.
Working directory flow:
1. Parse target commit and recursively load its tree.
2. For each tracked path, create, update, or delete files/folders to match target snapshot.
3. Recreate executable vs non-executable mode correctly.
4. Handle file-to-directory and directory-to-file conflicts carefully.
Why this is complex:
1. You must not overwrite uncommitted local changes.
2. Untracked files can block checkout if target branch needs that path.
3. File and directory replacement is order-sensitive.
4. There is no single atomic filesystem transaction for all changes, so operation ordering and failure handling matter to keep metadata, index, and working tree consistent.

Q5.2: Detecting dirty working directory conflicts
Use a three-way comparison per path:
1. Current commit vs index: tells staged changes.
2. Index vs working tree: tells unstaged changes.
3. Current commit vs target commit: tells what checkout will modify.
Conflict rule:
1. A path is locally dirty (staged or unstaged), and
2. Checkout would change that same path (or its type), and
3. Local content is not already identical to target content.
Then checkout must refuse to avoid data loss.
Also check obstruction by untracked paths:
1. If a path is untracked locally but target checkout needs to create/update it, refuse checkout.

Q5.3: Detached HEAD
In detached HEAD state:
1. HEAD points directly to a commit, not a branch name.
2. New commits are still created normally, but no branch pointer moves to reference them.
3. They are not immediately lost, but can become unreachable later.

Recovery:
1. Create a branch pointing to the detached commit (or latest detached commit).
2. Switch HEAD back to that branch so future commits remain referenced.
3. In full Git, reflog helps find recent detached commits; in a simple VCS, recovery is usually manual by known commit hash.

Q6.1: Finding unreachable objects
Use mark-and-sweep garbage collection.
Mark phase:
1. Start from all branch tip commits (and detached HEAD commit if applicable).
2. Traverse commit graph through parent links.
3. From each commit, traverse its tree recursively to visit subtrees and blobs.
4. Store visited hashes in a hash set for O(1) membership checks.
Sweep phase:
1. Walk all stored objects in object storage.
2. Delete objects not present in the reachable set.
Complexity estimate:
1. Work is proportional to number of unique reachable objects, not branches multiplied by commits.
2. For 100,000 commits and 50 branches with shared history, traversal is often near one full history plus unique side branches.
3. Reachable objects could be hundreds of thousands to a few million depending on how many unique trees/blobs were created.

Q6.2: Why GC and commit can race
Race condition:
1. Commit writes blobs/trees/commit object first.
2. Before branch ref update, these new objects may appear unreachable from refs.
3. Concurrent GC scans refs, does not see them as reachable, and deletes them.
4. Commit then updates branch to point to now-missing objects, causing corruption.
How real Git avoids this:
1. Uses reference locking/transactions around updates.
2. Uses conservative pruning (age-based grace period) instead of deleting very recent unreachable objects immediately.
3. Treats reflog entries as additional roots during retention windows.
4. Uses safe GC workflows so newly created objects are not pruned during active operations.